Jan. 20

- Main objects of study are groups and rings
- Algebra – the restoring of broken parts

  $x+5=7$ $\quad$ $2x^2-x=0$, used to working in $\mathbb{R}$

  - where else can we solve equations? → Groups & Rings
  - What do we need to solve equations? ↗ Groups & Rings
- Group – a set with one operation that behaves nicely
- Ring – a set with two operations that behave nicely
  - Very general places to solve equations

HW 0

1) $x \equiv 3 \mod 5$ → x and 3 have the same remainder when divided by 5. Equivalently, 5 divides $x-3$

   $x = 3, 8, 13, 18, 23 \ldots$ ← counting up by 5

   $= -2, -7, -12, \ldots$ ← counting down by 5

4) Simplify $7771 \cdot (-22) \mod 7$

   7 divides 7770, so 7771 has remainder 1 when divided by 7

   ↳ $7771 \equiv 1 \mod 7$

   So $7771 \cdot (-22) \equiv 1 \cdot (-22)$

   $\equiv 1 \cdot (-1) \equiv -1 \mod 7$

   or $\equiv 1 \cdot 6 \equiv 6 \mod 7$

   7 divides -21, so -22 has remainder -1 (also congr. 6)

ex. $6^{280} \mod 7 \equiv (-1)^{280} \mod 7$

also written $6^{280} \equiv (-1)^{280} \equiv 1 \mod 7$

Made with Goodnotes